

Complex Fractions

Name
Date
Course

Simplify.

$$\textcircled{1} \frac{\frac{5}{6}}{\frac{2}{3}} = \frac{5}{6} \div \frac{2}{3} = \frac{5}{6} \cdot \frac{3}{2} = \frac{15}{12} = \boxed{\frac{5}{4}}$$

$$\textcircled{2} \frac{\frac{7}{12}}{\frac{3}{8}} = \frac{7}{12} \div \frac{3}{8} = \frac{7}{12} \cdot \frac{8}{3} = \frac{8 \cdot 7}{12 \cdot 3} = \frac{2 \cdot 7}{3 \cdot 3} = \boxed{\frac{14}{9}}$$

$$\textcircled{3} \frac{-\frac{1}{7}}{\frac{4}{5}} = -\frac{1}{7} \div \frac{4}{5} = -\frac{1}{7} \cdot \frac{5}{4} = \boxed{-\frac{5}{28}}$$

$$\textcircled{4} \frac{-\frac{3}{4}}{-\frac{5}{8}} = -\frac{3}{4} \div \left(-\frac{5}{8}\right) = -\frac{3}{4} \cdot \left(-\frac{8}{5}\right) = \frac{24}{20} = \boxed{\frac{6}{5}}$$

$$\textcircled{5} \frac{\frac{4}{15}}{\frac{5}{6}} = \frac{4}{15} \div \frac{5}{6} = \frac{4}{15} \cdot \frac{6}{5} = \frac{24}{75} = \boxed{\frac{8}{25}}$$

$$\textcircled{6} \frac{\frac{1}{4}}{-\frac{3}{5}} = \frac{1}{4} \div \left(-\frac{3}{5}\right) = \frac{1}{4} \cdot \left(-\frac{5}{3}\right) = \boxed{-\frac{5}{12}}$$

$$\textcircled{7} \frac{\frac{2}{5x}}{\frac{3}{10}} = \frac{2}{5x} \div \frac{3}{10} = \frac{2}{5x} \cdot \frac{10}{3} = \frac{20}{15x} = \boxed{\frac{4}{3x}}$$

$$\textcircled{8} \frac{-\frac{2}{7ab^2}}{\frac{3}{a^2b}} = -\frac{2}{7ab^2} \div \frac{3}{a^2b} = -\frac{2}{7ab^2} \cdot \frac{a^2b}{3} = -\frac{2a^2b}{21ab^2} = \boxed{-\frac{2a}{21b}}$$

$$\textcircled{9} \frac{\frac{2}{9rt}}{\frac{1}{r^2t^2}} = \frac{2}{9rt} \div \frac{1}{r^2t^2} = \frac{2}{9rt} \cdot \frac{r^2t^2}{1} = \frac{2r^2t^2}{9rt} = \boxed{\frac{2rt}{9}}$$

$$\textcircled{10} \frac{\frac{2}{3}}{\frac{3}{6y^2}} = \frac{2}{3} \div \frac{3}{6y^2} = \frac{2}{3} \cdot \frac{6y^2}{3} = \frac{12y^2}{9} = \boxed{\frac{4y^2}{3}}$$

$$\textcircled{11} \frac{\frac{5}{6ab^2}}{\frac{2}{9ab}} = \frac{5}{6ab^2} \div \frac{2}{9ab} = \frac{5}{6ab^2} \cdot \frac{9ab}{2} = \frac{45ab}{12ab^2} = \boxed{\frac{15}{4b}}$$

$$\textcircled{12} \frac{-\frac{1}{4rt}}{-\frac{3}{r^2}} = -\frac{1}{4rt} \div \left(-\frac{3}{r^2}\right) = -\frac{1}{4rt} \cdot \left(-\frac{r^2}{3}\right) = \frac{r^2}{12rt} = \boxed{\frac{r}{12t}}$$

$$\textcircled{13} \frac{\frac{3}{x+2}}{\frac{7}{x}} = \frac{3}{x+2} \div \frac{7}{x} = \frac{3}{x+2} \cdot \frac{x}{7} = \boxed{\frac{3x}{7x+14}}$$

$$\textcircled{14} \quad \frac{\frac{5}{x}}{\frac{2}{x+4}} = \frac{5}{x} \div \frac{2}{x+4} = \frac{5}{x} \cdot \frac{x+4}{2} = \boxed{\frac{5x+20}{2x}}$$

$$\textcircled{15} \quad \frac{\frac{6}{5y+2}}{\frac{7}{5y^2+2y}} = \frac{6}{5y+2} \div \frac{7}{5y^2+2y} = \frac{6}{5y+2} \cdot \frac{5y^2+2y}{7} = \frac{6}{5y+2} \cdot \frac{y(5y+2)}{7} = \boxed{\frac{6y}{7}}$$

$$\textcircled{16} \quad \frac{\frac{3}{y}}{\frac{4}{y^2-7y}} = \frac{3}{y} \div \frac{4}{y^2-7y} = \frac{3}{y} \cdot \frac{y^2-7y}{4} = \frac{3}{y} \cdot \frac{y(y-7)}{4} = \boxed{\frac{3y-21}{4}}$$

$$\textcircled{17} \quad \frac{\frac{1}{2z^3-8z^2}}{\frac{2z}{(z-4)^2}} = \frac{1}{2z^3-8z^2} \div \frac{2z}{(z-4)^2} = \frac{1}{2z^3-8z^2} \cdot \frac{(z-4)^2}{2z} = \frac{1}{2z^2(z-4)} \cdot \frac{(z-4)^2}{2z} = \boxed{\frac{z-4}{4z^3}}$$

$$\textcircled{18} \quad \frac{\frac{4}{z^2+3z}}{\frac{2}{7z}} = \frac{4}{z^2+3z} \div \frac{2}{7z} = \frac{4}{z^2+3z} \cdot \frac{7z}{2} = \frac{4}{z(z+3)} \cdot \frac{7z}{2} = \frac{28}{2(z+3)} = \boxed{\frac{14}{z+3}}$$

$$\textcircled{19} \quad \frac{\frac{2}{x+3}}{\frac{5}{x^2-4x-21}} = \frac{2}{x+3} \div \frac{5}{x^2-4x-21} = \frac{2}{x+3} \cdot \frac{x^2-4x-21}{5} = \frac{2}{x+3} \cdot \frac{(x-7)(x+3)}{5} = \boxed{\frac{2x-14}{5}}$$

$$\textcircled{20} \quad \frac{\frac{2}{x^2+5x+4}}{\frac{3}{x+1}} = \frac{2}{x^2+5x+4} \div \frac{3}{x+1} = \frac{2}{x^2+5x+4} \cdot \frac{x+1}{3} = \frac{2}{(x+4)(x+1)} \cdot \frac{x+1}{3} = \boxed{\frac{2}{3x+12}}$$

$$\textcircled{21} \quad \frac{\frac{6}{y^2-25}}{\frac{11}{y+5}} = \frac{6}{y^2-25} \div \frac{11}{y+5} = \frac{6}{y^2-25} \cdot \frac{y+5}{11} = \frac{6}{(y-5)(y+5)} \cdot \frac{y+5}{11} = \boxed{\frac{6}{11y-55}}$$

$$\begin{aligned} \textcircled{22} \quad \frac{\frac{8}{y+3}}{\frac{2}{y^2-9}} &= \frac{8}{y+3} \div \frac{2}{y^2-9} = \frac{8}{y+3} \cdot \frac{y^2-9}{2} = \frac{8}{y+3} \cdot \frac{(y-3)(y+3)}{2} \\ &= \frac{8y-24}{2} \\ &= \boxed{4y-12} \end{aligned}$$

$$\begin{aligned} \textcircled{23} \quad \frac{\frac{3}{4} + \frac{1}{6}}{\frac{5}{6}} &= \frac{\frac{9}{12} + \frac{2}{12}}{\frac{5}{6}} = \frac{\frac{11}{12}}{\frac{5}{6}} = \frac{11}{12} \div \frac{5}{6} \\ &= \frac{11}{12} \cdot \frac{6}{5} = \frac{66}{60} \\ &= \boxed{\frac{11}{10}} \end{aligned}$$

$$\begin{aligned} \textcircled{24} \quad \frac{\frac{7}{25} + \frac{11}{50}}{\frac{3}{10}} &= \frac{\frac{14}{50} + \frac{11}{50}}{\frac{3}{10}} = \frac{\frac{25}{50}}{\frac{3}{10}} = \frac{\frac{1}{2}}{\frac{3}{10}} = \frac{1}{2} \div \frac{3}{10} \\ &= \frac{1}{2} \cdot \frac{10}{3} \\ &= \frac{10}{6} \\ &= \boxed{\frac{5}{3}} \end{aligned}$$

$$\begin{aligned} \textcircled{25} \quad \frac{\frac{2}{3}}{\frac{5}{6} - \frac{1}{2}} &= \frac{\frac{2}{3}}{\frac{5}{6} - \frac{3}{6}} = \frac{\frac{2}{3}}{\frac{2}{6}} = \frac{\frac{2}{3}}{\frac{1}{3}} = \frac{2}{3} \div \frac{1}{3} \\ &= \frac{2}{3} \cdot \frac{3}{1} \\ &= \frac{6}{3} \\ &= \boxed{2} \end{aligned}$$

$$\begin{aligned} \textcircled{26} \quad \frac{\frac{5}{6}}{\frac{5}{12} - \frac{1}{3}} &= \frac{\frac{5}{6}}{\frac{5}{12} - \frac{4}{12}} = \frac{\frac{5}{6}}{\frac{1}{12}} = \frac{5}{6} \div \frac{1}{12} = \frac{5}{6} \cdot \frac{12}{1} \\ &= \frac{60}{6} \\ &= \boxed{10} \end{aligned}$$

$$\begin{aligned} \textcircled{27} \quad \frac{\frac{1}{6} - \frac{5}{8}}{\frac{1}{2} - \frac{1}{3}} &= \frac{\frac{4}{24} - \frac{15}{24}}{\frac{3}{6} - \frac{2}{6}} = \frac{\frac{-11}{24}}{\frac{1}{6}} = \frac{-11}{24} \div \frac{1}{6} \\ &= \frac{-11}{24} \cdot \frac{6}{1} \\ &= -\frac{66}{24} \\ &= \boxed{-\frac{11}{4}} \end{aligned}$$

$$\begin{aligned} \textcircled{28} \quad \frac{\frac{3}{4} + \frac{5}{8}}{\frac{1}{2} - \frac{9}{10}} &= \frac{\frac{6}{8} + \frac{5}{8}}{\frac{5}{10} - \frac{9}{10}} = \frac{\frac{11}{8}}{-\frac{4}{10}} = \frac{11}{8} \div \left(-\frac{4}{10}\right) \\ &= \frac{11}{8} \cdot \left(-\frac{10}{4}\right) \\ &= \frac{11}{8} \cdot \left(-\frac{5}{2}\right) \\ &= \boxed{-\frac{55}{16}} \end{aligned}$$

$$\begin{aligned} \textcircled{29} \quad \frac{\frac{1}{4x} + \frac{5}{8}}{\frac{1}{2} - \frac{1}{4x^2}} &= \frac{\frac{2}{8x} + \frac{5x}{8x}}{\frac{2x^2}{4x^2} - \frac{1}{4x^2}} = \frac{\frac{2+5x}{8x}}{\frac{2x^2-1}{4x^2}} = \frac{2+5x}{8x} \cdot \frac{2x^2-1}{4x^2} \\ &= \frac{2+5x}{8x} \cdot \frac{4x^2}{2x^2-1} \\ &= \frac{4x^2(2+5x)}{8x(2x^2-1)} \\ &= \boxed{\frac{x(2+5x)}{2(2x^2-1)}} \end{aligned}$$

$$\begin{aligned} \textcircled{30} \quad \frac{\frac{1}{9x^5} - \frac{2}{3}}{\frac{7}{9x} - \frac{2}{3x^2}} &= \frac{\frac{1}{9x^5} - \frac{6x^5}{9x^5}}{\frac{7x}{9x^2} - \frac{6}{9x^2}} = \frac{\frac{1-6x^5}{9x^5}}{\frac{7x-6}{9x^2}} = \frac{1-6x^5}{9x^5} \cdot \frac{9x^2}{7x-6} \\ &= \frac{1-6x^5}{9x^5} \cdot \frac{9x^2}{7x-6} \end{aligned}$$

$$\begin{aligned} &= \frac{1-6x^5}{x^3(7x-6)} \\ &= \boxed{\frac{1-6x^5}{7x^4-6x^3}} \end{aligned}$$

$$\begin{aligned} \textcircled{31} \quad \frac{\frac{8}{y-2}}{\frac{5}{y^2+y-6} + \frac{7}{y+3}} &= \frac{\frac{8}{y-2}}{\frac{5}{(y+3)(y-2)} + \frac{7}{y+3}} \\ &= \frac{\frac{8}{y-2}}{\frac{5}{(y+3)(y-2)} + \frac{7(y-2)}{(y+3)(y-2)}} \end{aligned}$$

$$\begin{aligned} &= \frac{\frac{8}{y-2}}{\frac{5}{(y+3)(y-2)} + \frac{7y-14}{(y+3)(y-2)}} \\ &= \frac{\frac{8}{y-2}}{\frac{7y-9}{(y+3)(y-2)}} \end{aligned}$$

$$\begin{aligned} &= \frac{\frac{8}{y-2}}{\frac{7y-9}{(y+3)(y-2)}} \\ &= \frac{8}{y-2} \div \frac{7y-9}{(y+3)(y-2)} \end{aligned}$$

$$\begin{aligned} &= \frac{8}{y-2} \cdot \frac{(y+3)(y-2)}{7y-9} \\ &= \boxed{\frac{8y+24}{7y-9}} \end{aligned}$$

$$\textcircled{32} \quad \frac{\frac{8}{y^2-8y-20}}{\frac{5}{y+2} + \frac{7}{y-10}}$$

$$\begin{aligned} &= \frac{\frac{8}{(y+2)(y-10)}}{\frac{5}{y+2} + \frac{7}{y-10}} \end{aligned}$$

$$\begin{aligned} &= \frac{\frac{8}{(y+2)(y-10)}}{\frac{(y-10)5}{(y-10)(y+2)} + \frac{7(y+2)}{(y-10)(y+2)}} \end{aligned}$$

$$\begin{aligned} &= \frac{\frac{8}{(y+2)(y-10)}}{\frac{5y-50}{(y-10)(y+2)} + \frac{7y+14}{(y-10)(y+2)}} \end{aligned}$$

$$\begin{aligned} &= \frac{\frac{8}{(y+2)(y-10)}}{\frac{12y-36}{(y-10)(y+2)}} \end{aligned}$$

$$\begin{aligned} &= \frac{8}{(y+2)(y-10)} \div \frac{12y-36}{(y-10)(y+2)} \end{aligned}$$

$$\begin{aligned} &= \frac{8}{(y+2)(y-10)} \cdot \frac{(y-10)(y+2)}{12y-36} \end{aligned}$$

$$\begin{aligned} &= \frac{8}{12y-36} \end{aligned}$$

$$\boxed{\frac{2}{3y-9}}$$

$$\textcircled{33} \quad \frac{\frac{3}{z} - \frac{1}{z-6}}{\frac{7}{z^2-6z}}$$

$$\frac{\frac{3}{z} - \frac{1}{z-6}}{\frac{7}{z(z-6)}}$$

$$\frac{\frac{(z-6)3}{(z-6)z} - \frac{1}{(z-6)(z)}}{\frac{7}{z(z-6)}}$$

$$\frac{\frac{3z-18}{z(z-6)} - \frac{z}{z(z-6)}}{\frac{7}{z(z-6)}}$$

$$\frac{\frac{2z-18}{z(z-6)}}{\frac{7}{z(z-6)}}$$

$$\frac{2z-18}{z(z-6)} \cdot \frac{7}{z(z-6)}$$

$$\frac{2z-18}{z(z-6)} \cdot \frac{z(z-6)}{7}$$

$$\boxed{\frac{2z-18}{7}}$$

$$\textcircled{34} \quad \frac{\frac{4}{z} - \frac{2}{z^2-2z}}{\frac{3}{z-2}}$$

$$\frac{\frac{4}{z} - \frac{2}{z(z-2)}}{\frac{3}{z-2}}$$

$$\frac{\frac{(z-2)4}{(z-2)z} - \frac{2}{z(z-2)}}{\frac{3}{z-2}}$$

$$\frac{\frac{4z-8}{z(z-2)} - \frac{2}{z(z-2)}}{\frac{3}{z-2}}$$

$$\frac{\frac{4z-10}{z(z-2)}}{\frac{3}{z-2}}$$

$$\frac{4z-10}{z(z-2)} \cdot \frac{3}{z-2}$$

$$\frac{4z-10}{z(z-2)} \cdot \frac{z-2}{3}$$

$$\boxed{\frac{4z-10}{3z}}$$

$$\textcircled{35} \quad \frac{\frac{2}{x+7} + \frac{5}{x^2-49}}{\frac{7}{x-7} + \frac{8}{x+7}}$$

$$\frac{\frac{2}{x+7} + \frac{5}{(x-7)(x+7)}}{\frac{7}{x-7} + \frac{8}{x+7}}$$

$$\frac{\frac{(x-7)2}{(x-7)(x+7)} + \frac{5}{(x-7)(x+7)}}{\frac{(x+7)7}{(x+7)(x-7)} + \frac{8(x-7)}{(x+7)(x-7)}}$$

$$\frac{\frac{2x-14}{(x-7)(x+7)} + \frac{5}{(x-7)(x+7)}}{\frac{7x+49}{(x+7)(x-7)} + \frac{8x-56}{(x+7)(x-7)}}$$

$$\frac{\frac{2x-9}{(x-7)(x+7)}}{\frac{15x-7}{(x+7)(x-7)}}$$

$$\frac{2x-9}{(x-7)(x+7)} \div \frac{15x-7}{(x+7)(x-7)}$$

$$\frac{2x-9}{(x-7)(x+7)} \cdot \frac{(x+7)(x-7)}{15x-7}$$

$$\boxed{\frac{2x-9}{15x-7}}$$

$$\textcircled{36} \quad \frac{\frac{2}{x-6} + \frac{5}{x+6}}{\frac{7}{x^2-36} + \frac{8}{x+6}}$$

$$\frac{\frac{2}{x-6} + \frac{5}{x+6}}{\frac{7}{(x-6)(x+6)} + \frac{8}{x+6}}$$

$$\frac{\frac{(x+6)2}{(x+6)(x-6)} + \frac{5(x-6)}{(x+6)(x-6)}}{\frac{7}{(x-6)(x+6)} + \frac{8(x-6)}{(x+6)(x-6)}}$$

$$\frac{\frac{2x+12}{(x-6)(x+6)} + \frac{5x-30}{(x+6)(x-6)}}{\frac{7}{(x-6)(x+6)} + \frac{8x-48}{(x+6)(x-6)}}$$

$$\frac{\frac{7x-18}{(x-6)(x+6)}}{\frac{8x-41}{(x-6)(x+6)}}$$

$$\frac{7x-18}{(x-6)(x+6)} \div \frac{8x-41}{(x-6)(x+6)}$$

$$\frac{7x-18}{(x-6)(x+6)} \cdot \frac{(x-6)(x+6)}{8x-41}$$

$$\boxed{\frac{7x-18}{8x-41}}$$

$$\textcircled{37} \quad \frac{2}{y^2-11y+24} + \frac{5}{y-8}$$

$$\frac{4}{y-8} + \frac{2}{y-3}$$

$$\frac{2}{(y-3)(y-8)} + \frac{5}{y-8}$$

$$\frac{4}{y-8} + \frac{2}{y-3}$$

$$\frac{2}{(y-3)(y-8)} + \frac{5(y-3)}{(y-8)(y-3)}$$

$$\frac{(y-3)4}{(y-3)(y-8)} + \frac{2(y-8)}{(y-3)(y-8)}$$

$$\frac{2}{(y-3)(y-8)} + \frac{5y-15}{(y-8)(y-3)}$$

$$\frac{4y-12}{(y-3)(y-8)} + \frac{2y-16}{(y-3)(y-8)}$$

$$\frac{5y-13}{(y-3)(y-8)}$$

$$\frac{6y-28}{(y-3)(y-8)}$$

$$\frac{5y-13}{(y-3)(y-8)} \div \frac{6y-28}{(y-3)(y-8)}$$

$$\frac{5y-13}{(y-3)(y-8)} \cdot \frac{(y-3)(y-8)}{6y-28}$$

$$\boxed{\frac{5y-13}{6y-28}}$$

$$\textcircled{38} \quad \frac{1}{y-2} + \frac{-3}{y+4}$$

$$\frac{2}{y^2+2y-8} - \frac{6}{y+4}$$

$$\frac{1}{y-2} + \frac{-3}{y+4}$$

$$\frac{2}{(y+4)(y-2)} - \frac{6}{y+4}$$

$$\frac{(y+4)1}{(y+4)(y-2)} + \frac{-3(y-2)}{(y+4)(y-2)}$$

$$\frac{2}{(y+4)(y-2)} - \frac{6(y-2)}{(y+4)(y-2)}$$

$$\frac{y+4}{(y+4)(y-2)} + \frac{-3y+6}{(y+4)(y-2)}$$

$$\frac{2}{(y+4)(y-2)} - \frac{6y-12}{(y+4)(y-2)}$$

$$\frac{-2y+10}{(y+4)(y-2)}$$

$$\frac{-6y+14}{(y+4)(y-2)}$$

$$\frac{-2y+10}{(y+4)(y-2)} \div \frac{-6y+14}{(y+4)(y-2)}$$

$$\frac{-2y+10}{(y+4)(y-2)} \cdot \frac{(y+4)(y-2)}{-6y+14}$$

$$\frac{-2y+10}{-6y+14}$$

or

$$\frac{-y+5}{-3y+7}$$

$$\frac{y-5}{3y-7}$$

$$\textcircled{39} \quad \frac{2}{z} + \frac{5}{z^2-2z}$$

$$\frac{1}{z-2} - \frac{3}{z}$$

$$\textcircled{40} \quad \frac{2}{z+8} - \frac{8}{3}$$

$$\frac{2}{3} - \frac{5}{3z^2+24z}$$

$$\textcircled{41} \quad \frac{3 + \frac{2}{x+1}}{1 - \frac{3}{x+1}}$$

$$\frac{\frac{3}{1} + \frac{2}{x+1}}{1 - \frac{3}{x+1}}$$

$$\frac{\frac{(x+1)3}{(x+1)1} + \frac{2}{x+1}}{\frac{(x+1)1}{(x+1)1} - \frac{3}{x+1}}$$

$$\frac{\frac{3x+3}{x+1} + \frac{2}{x+1}}{\frac{x+1}{x+1} - \frac{3}{x+1}}$$

$$\frac{\frac{3x+5}{x+1}}{\frac{x-2}{x+1}}$$

$$\frac{3x+5}{x+1} \div \frac{x-2}{x+1}$$

$$\frac{3x+5}{x+1} \cdot \frac{x+1}{x-2}$$

$$\boxed{\frac{3x+5}{x-2}}$$

$$\textcircled{42} \quad \frac{\frac{3}{x-2} + 7}{1 - \frac{3}{x-2}}$$

$$\textcircled{43} \quad \frac{\frac{a}{a+2} - a^2}{5a - \frac{3}{a+2}}$$

$$\frac{\frac{a}{a+2} - \frac{a^2}{1}}{5a - \frac{3}{a+2}}$$

$$\frac{\frac{a}{a+2} - \frac{a^2(a+2)}{1(a+2)}}{(a+2)5a - \frac{3}{a+2}}$$

$$\frac{(a+2)5a - \frac{3}{a+2}}{(a+2)1 - \frac{3}{a+2}}$$

$$\frac{\frac{a}{a+2} - \frac{a^3+2a^2}{a+2}}{\frac{5a^2+10a}{a+2} - \frac{3}{a+2}}$$

$$\frac{\frac{a-a^3-2a^2}{a+2}}{\frac{5a^2+10a-3}{a+2}}$$

$$\frac{a-a^3-2a^2}{a+2} \div \frac{5a^2+10a-3}{a+2}$$

$$\frac{a-a^3-2a^2}{a+2} \cdot \frac{a+2}{5a^2+10a-3}$$

$$\boxed{\frac{a-a^3-2a^2}{5a^2+10a-3}}$$

$$\textcircled{44} \quad \frac{\frac{a}{a+4} + 3a^5}{\frac{3}{a+4} - 2a}$$

$$\textcircled{45} \quad \frac{2}{y+2} - \frac{5}{3y^2+8y+4}$$

$$9 + \frac{1}{y+2}$$

$$\frac{2}{y+2} - \frac{5}{(3y+2)(y+2)}$$

$$\frac{9}{1} + \frac{1}{y+2}$$

$$\frac{(3y+2)2}{(3y+2)(y+2)} - \frac{5}{(3y+2)(y+2)}$$

$$\frac{(3y+2)(y+2)9}{(3y+2)(y+2)1} + \frac{1}{(y+2)(3y+2)}$$

$$\frac{6y+4}{(3y+2)(y+2)} - \frac{5}{(3y+2)(y+2)}$$

$$\frac{(3y^2+8y+4)(9)}{(3y+2)(y+2)} + \frac{3y+2}{(y+2)(3y+2)}$$

$$\frac{6y+4}{(3y+2)(y+2)} - \frac{5}{(3y+2)(y+2)}$$

$$\frac{27y^2+72y+36}{(3y+2)(y+2)} + \frac{3y+2}{(y+2)(3y+2)}$$

$$\frac{6y-1}{(3y+2)(y+2)}$$

$$\frac{27y^2+75y+38}{(y+2)(3y+2)}$$

$$\frac{6y-1}{(3y+2)(y+2)} \div \frac{27y^2+75y+38}{(y+2)(3y+2)}$$

$$\frac{6y-1}{(3y+2)(y+2)} \cdot \frac{(y+2)(3y+2)}{27y^2+75y+38}$$

$$\boxed{\frac{6y-1}{27y^2+75y+38}}$$

$$\textcircled{46} \quad \frac{2}{2y^2+5y-12} - \frac{5}{2y-3}$$

$$\frac{2}{y+4} + 2$$

$$\textcircled{47} \quad \frac{2}{\frac{3}{4}} = 2 \div \frac{3}{4} = \frac{2}{1} \div \frac{3}{4} = \frac{2}{1} \cdot \frac{4}{3} = \boxed{\frac{8}{3}}$$

$$\textcircled{48} \quad \frac{-\frac{5}{6}}{8} = -\frac{5}{6} \div 8 = -\frac{5}{6} \div \frac{8}{1} = -\frac{5}{6} \cdot \frac{1}{8} = \boxed{-\frac{5}{48}}$$

$$\textcircled{49} \quad \frac{6x}{\frac{3x}{2}} = 6x \div \frac{3x}{2} = \frac{6x}{1} \div \frac{3x}{2} = \frac{6x}{1} \cdot \frac{2}{3x} = \frac{12x}{3x} = \boxed{4}$$

$$\textcircled{50} \quad \frac{\frac{2y}{5}}{-7y} = \frac{2y}{5} \div (-7y) = \frac{2y}{5} \div \left(\frac{-7y}{1}\right)$$

$$= \frac{2y}{5} \div \left(-\frac{1}{7y}\right)$$

$$= -\frac{2y}{35y}$$

$$= \boxed{-\frac{2}{35}}$$